

5-1

(1) $p = 20 \text{ MPa}$, $t = 300^\circ\text{C}$

$t_s(20 \text{ MPa}) = 365.789^\circ\text{C} > 300^\circ\text{C}$ 故该点为未饱和水

未饱和水 300°C , $v = 0.001365 \text{ m}^3/\text{kg}$

$h = 1333.4 \text{ kJ/kg}$

$s = 3.2072 \text{ kJ/kg}\cdot\text{K}$

(2) $p = 9 \text{ MPa}$, $v = 0.017 \text{ m}^3/\text{kg}$

$v' = 0.0014177 \text{ m}^3/\text{kg} < v < v'' = 0.020485 \text{ m}^3/\text{kg}$

$\Rightarrow$ 该点为湿饱和蒸汽

$x = \frac{v - v'}{v'' - v'} = \frac{0.017 - 0.0014177}{0.020485 - 0.0014177} = 0.8172$

$h_x = xh'' + (1-x)h' = 0.8172(2741.92) + (1-0.8172)(1363.1)$
 $= 2489.8717 \text{ kJ/kg}$

$s_x = xs'' + (1-x)s' = 0.8172(5.6771) + (1-0.8172)(3.2854)$
 $= 5.2399 \text{ kJ/kg}\cdot\text{K}$

$t = 300^\circ\text{C} + \frac{9 - 8.58308}{9.85970 - 8.58308} \times 10 = 303.2658^\circ \approx 303.27^\circ$

(3) $p = 4.5 \text{ MPa}$, $t = 450^\circ\text{C}$

当 $p = 5.0 \text{ MPa}$, $t_s(5.0 \text{ MPa}) = 263.980^\circ\text{C} < t \Rightarrow$ 该点为过热蒸汽

450°C , 3 MPa $v = 0.107864 \text{ m}^3/\text{kg}$

450°C , 5 MPa $v = 0.063291 \text{ m}^3/\text{kg}$

450°C , 4.5 MPa $v = 0.107864 - \frac{1.5}{2}(0.044573) = 0.0744 \text{ m}^3/\text{kg}$

$$450^{\circ}\text{C}, 3\text{MPa} \quad s = 7.0817 \text{ kJ/kg}\cdot\text{K}$$

$$450^{\circ}\text{C}, 5\text{MPa} \quad s = 6.8170 \text{ kJ/kg}\cdot\text{K}$$

$$450^{\circ}\text{C}, 4.5\text{MPa} \quad s = 7.0817 - \frac{1.5}{2}(0.2647) = \underline{6.8832 \text{ kJ/kg}\cdot\text{K}}$$

$$450^{\circ}\text{C}, 3\text{MPa} \quad h = 3343.0 \text{ kJ/kg}$$

$$450^{\circ}\text{C}, 5\text{MPa} \quad h = 3315.2 \text{ kJ/kg}$$

$$450^{\circ}\text{C}, 4.5\text{MPa} \quad h = 3343 - \frac{1.5}{2}(27.8) = \underline{3322.15 \text{ kJ/kg}}$$

$$(4) \quad p = 1 \text{ MPa}, x = 0.9$$

$$x = \frac{v - v'}{v'' - v'} \quad v = v' + x(v'' - v') = 0.0011272 + 0.9(0.19434 - 0.0011272)$$

$$v = \underline{0.17502 \text{ m}^3/\text{kg}}$$

$$s = xs'' + (1-x)s' = 0.9(6.5859) + 0.1(2.1388) = \underline{6.14119 \text{ kJ/kg}\cdot\text{K}}$$

$$h = xh'' + (1-x)h' = 0.9(2777.67) + 0.1(762.84) = \underline{2576.187 \text{ kJ/kg}}$$

$$t = 170 + \frac{1 - 0.79147}{1.00193 - 0.79147} (10) = \underline{179.9083^{\circ}\text{C}} < t_s = 179.916^{\circ}\text{C}$$

$\Rightarrow$ 该点为未饱和水

5-3

$$0.35\text{MPa} \quad V = 1\text{m}^3 \quad \text{饱和蒸汽} \quad h_1 = 2732.37 \text{ kJ/kg}$$

$$v'' = 0.52427 \text{ m}^3/\text{kg} \quad \text{干饱和蒸汽质量为 } \frac{1}{0.52427} = \underline{1.9074 \text{ kg}}$$

$$vM = V$$

$$m = \frac{V}{v}$$

$$0.35\text{MPa} \rightarrow 0.2\text{MPa}, \quad v_{\text{不变}} = 0.52427 \text{ m}^3/\text{kg}$$

$$v' = 0.0010605 \text{ m}^3/\text{kg} < v < v'' = 0.88585 \text{ m}^3/\text{kg} \Rightarrow \text{湿饱和蒸汽}$$

$$x = \frac{v - v'}{v'' - v'} = \frac{0.52427 - 0.0010605}{0.88585 - 0.0010605} = 0.5913 \approx 0.59$$

$$h_2 = xh'' + (1-x)h' = 0.5913(2706.53) + (1-0.5913)(504.78) \\ = 1806.6748 \text{ kJ/kg}$$

$$Q = m(h_1 - h_2) = 1.9074(2732.37 - 1806.6748) = \underline{1765.6710 \text{ kJ}}$$

5-7

$$p = 1.5 \text{ MPa}, \quad x = 0.95$$

$$h' = 844.82 \text{ kJ/kg} \quad h'' = 2791.46 \text{ kJ/kg}$$

$$s' = 2.3149 \text{ kJ/kg}\cdot\text{K} \quad s'' = 6.4437 \text{ kJ/kg}\cdot\text{K}$$

$$v' = 0.001538 \text{ m}^3/\text{kg} \quad v'' = 0.13172 \text{ m}^3/\text{kg}$$

$$t_s(1.5 \text{ MPa}) = 198.327^\circ\text{C}$$

$$h = xh'' + (1-x)h' = 0.95(2791.46) + 0.05(844.82) = 2684.128 \text{ kJ/kg}$$

$$p_2 = 0.025 \text{ MPa}, \quad s' = 0.4761 \text{ kJ/kg}\cdot\text{K} \quad s'' = 8.3930 \text{ kJ/kg}\cdot\text{K}$$

$$\text{蒸汽经过绝热膨胀, 熵不变} \Rightarrow s_1 = s_2 = xs'' + (1-x)s'$$

$$= 0.9(8.393) + 0.1(0.4761)$$

$$= 7.6013 \text{ kJ/kg}\cdot\text{K}$$

$$t = 440 + \frac{7.6013 - 7.5866}{7.6163 - 7.5866} (10) = 444.9495^\circ\text{C}$$

$$440^\circ\text{C}, 3 \text{ MPa}, \quad h = 3320.5 \text{ kJ/kg}$$

$$450^\circ\text{C}, 3 \text{ MPa}, \quad h = 3343.0 \text{ kJ/kg}$$

$$444.9495, 3 \text{ MPa}, \quad h = 3320.5 + \frac{444.9495 - 440}{450 - 440} (22.5) = 3331.6364 \text{ kJ/kg}$$

$$440^{\circ}\text{C}, 1\text{ MPa},$$

$$h = 3348.2 \text{ kJ/kg}$$

$$450^{\circ}\text{C}, 1\text{ MPa},$$

$$h = 3369.6 \text{ kJ/kg}$$

$$444.9495, 1\text{ MPa},$$

$$h = 3348.2 + \frac{444.9495 - 440}{450 - 440} (21.4) = 3358.7919 \text{ kJ/kg}$$

$$444.9495, 1.5\text{ MPa},$$

$$h_1 = 3358.7919 - \frac{1.5-1}{3-1} (27.1511) = 3352.003 \text{ kJ/kg}$$

$$Q = 1 \cdot (h_1 - h) = 3352.003 - 2694.128 = \underline{657.875 \text{ kJ/kg}}$$

5-8

湿饱和蒸汽

$$p = 0.4\text{ MPa}, \quad x = 0.04$$

$$m = 8250 \text{ kg}$$

$$Q = 17000 \text{ kJ/min}$$

$$v' = 0.0010835 \text{ m}^3/\text{kg}$$

$$v'' = 0.46246 \text{ m}^3/\text{kg}$$

$$h' = 604.87 \text{ kJ/kg}$$

$$h'' = 2738.49 \text{ kJ/kg}$$

$$s' = 1.7769 \text{ kJ/kg}\cdot\text{K}$$

$$s'' = 6.8961 \text{ kJ/kg}\cdot\text{K}$$

$$v_1 = xv'' + (1-x)v' = 0.04(0.46246) + 0.96(0.0010835) \\ = 0.01954 \text{ m}^3/\text{kg}$$

$$h_1 = xh'' + (1-x)h' = 0.04(2738.49) + 0.96(604.87) = 690.2148 \text{ kJ/kg}$$

$$u_1 = h_1 - p_1 v_1 = 690.2148 - \frac{0.4 \times 10^6}{1 \times 10^3} (0.01954) = 682.3988 \text{ kJ/kg}$$

$$p_2 = 1\text{ MPa}, \quad h_2 = h'' = 2777.67 \text{ kJ/kg} \quad v'' = 0.19438 \text{ m}^3/\text{kg}$$

$$u_2 = h_2 - p_2 v_2 = 2777.67 - 1 \times 10^3 (0.19438) = 2583.29 \text{ kJ/kg}$$

$$Q = m(u_2 - u_1) = 8250(2583.29 - 682.3988) = 1.5682 \times 10^7 \text{ kJ}$$

$$t = \frac{Q}{1.7 \times 10^4 / \text{min}} = 922.4706 \text{ min} \approx \underline{922 \text{ min}}$$